

UNIVERSITY OF CAPE TOWN

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II

Advanced Calculus (2AC)

VECTOR CALCULUS II: Line integrals of vector fields

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0.1 The line integral of a vector field

Let $\vec{F} : D \subset \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a vector field defined by

$$\vec{F}(x, y, z) = (P(x, y, z), Q(x, y, z), R(x, y, z)) \quad (\text{to be thought of as a force}) \quad \forall (x, y, z) \in D.$$

Before we give the definition of the line integral of a vector field $\vec{F}$ along a regular curve $\mathcal{C}$ from P_0 to P_1 , let us first consider the physical problem of finding the work done to move an object by the force $\vec{F}$.

If the vector field $\vec{F} = (\alpha, \beta, \delta)$ (constant force) and we move the object along the segment line from a point $P_0 = (a, b, c)$ to a point $P_1 = (a + \Delta x, b + \Delta y, c + \Delta z)$, then the work done is known to

$$W = \vec{F} \cdot \overrightarrow{P_0 P_1} \quad (\text{Dot product}).$$

Now, the force $\vec{F}$ may vary from point to point and the curve $\mathcal{C}$ need not to be a straight line. Precisely, how do we move from the elementary case we had before to a more sophisticated one. The approach is to proceed by approximation as follows. We consider points on the curve $\mathcal{C}$ as

$$Q_0 = P_0, Q_1, \dots, Q_n = P_1$$

and we consider the polygonal line obtained by joining the points Q_i by segment lines. Now, if we assume $\vec{F}$ is constant on each segment line i.e., $\vec{F}(x, y, z) = \vec{F}(x_i^*, y_i^*, z_i^*)$ for every $(x, y, z) \in Q_{i-1}Q_i$ and for some sample point (x_i^*, y_i^*, z_i^*) in the arc $Q_{i-1}Q_i$, then as an approximation of the work done W to the object along the curve $\mathcal{C}$ from the point P_0 to P_1 , we get

$$W \approx \sum_{i=1}^n \vec{F}(x_i^*, y_i^*, z_i^*) \cdot \overrightarrow{Q_{i-1}Q_i} \approx \sum_{i=1}^n \vec{F}(x_i^*, y_i^*, z_i^*) \cdot \vec{T}(x_i^*, y_i^*, z_i^*) \Delta s_i \quad (1)$$

where $\vec{T}(x_i^*, y_i^*, z_i^*)$ denotes the unit tangent vector to the curve $\mathcal{C}$ at the sample point (x_i^*, y_i^*, z_i^*) . Now, one would expect under reasonable conditions that the approximation improves as the number n is taken larger and the points Q_i closer together on $\mathcal{C}$ so that the exact expression for W would be as a kind of limit similar to those used to define the Riemann single and multiple integrals.

Notation:

$$W = \int_{\mathcal{C}} \vec{F} \cdot \vec{T} ds = \int_{\mathcal{C}} \vec{F} \cdot d\vec{r} \quad (2)$$

Going back to (2), we get that

$$\sum_{i=1}^n \vec{F}(x_i^*, y_i^*, z_i^*) \cdot \vec{T}(x_i^*, y_i^*, z_i^*) \Delta s_i \approx \sum_{i=1}^n \vec{F}(\vec{r}(t_i^*)) \cdot \vec{T}(t_i^*) \|\vec{r}'(t_i^*)\| \Delta t_i = \sum_{i=1}^n \vec{F}(\vec{r}(t_i^*)) \cdot \vec{r}'(t_i^*) \Delta t_i,$$

and hence, we get that

$$\boxed{W = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt \quad \text{where } P_0 = \vec{r}(a) \quad \text{and} \quad P_1 = \vec{r}(b)} \quad (3)$$

Definition 0.1 Let $\vec{F} : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a continuous vector field and let $\mathcal{C}$ be a curve described by the vector function $\vec{r} : [a, b] \rightarrow \mathbb{R}^n$ such that $\vec{r}(t) \in D$ for every $t \in [a, b]$. Then the line integral of the vector field $\vec{F}$ along the curve $\mathcal{C}$ is defined as

$$\boxed{\int_{\mathcal{C}} \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt = \int_{\mathcal{C}} \vec{F} \cdot \vec{T} ds} \quad (4)$$

Notation:

$$\int_{\mathcal{C}} \vec{F} \cdot d\vec{r} = \underbrace{\int_{\mathcal{C}} P dx + Q dy + R dz}_{\text{Differential notation}} \quad \text{in the case } n = 3.$$

Example 0.2 Evaluate the line integral $\int_{\mathcal{C}} \vec{F} \cdot d\vec{r}$ where $F(x, y, z) := (x, -y^2, xz)$ and $\mathcal{C}$ is the curve described by the vector function

$$\vec{r}(t) := (-t^2, 2t + 1, t^2 - 1) \quad -1 \leq t \leq 1.$$

Solution: $\vec{r}'(t) = (-2t, 2, 2t)$. Hence,

$$\begin{aligned} \int_{\mathcal{C}} \vec{F} \cdot d\vec{r} &= \int_{-1}^1 (-t^2, -(2t+1)^2, -t^2(t^2-1)) \cdot (-2t, 2, 2t) dt = \int_{-1}^1 [2t^3 - 2(2t+1)^2 - 2t^3(t^2-1)] dt \\ &= \int_{-1}^1 [-2t^5 + 4t^3 - 2(2t+1)^2] dt = \left[-\frac{1}{3}t^6 + t^4 - \frac{1}{3}(2t+1)^3 \right]_{-1}^1 = -\frac{28}{3}. \end{aligned}$$

Notice that if the vector field is parallel to the axes we get

$$\boxed{\begin{cases} \int_{\mathcal{C}} \vec{F} \cdot d\vec{r} = \int_{\mathcal{C}} P(x, y, z) dx = \int_a^b P(\vec{r}(t)) \frac{dx}{dt} dt & \text{if } \vec{F} = (P, 0, 0) \text{ i.e., parallel to the } x\text{-axis} \\ \int_{\mathcal{C}} \vec{F} \cdot d\vec{r} = \int_{\mathcal{C}} Q(x, y, z) dy = \int_a^b Q(\vec{r}(t)) \frac{dy}{dt} dt & \text{if } \vec{F} = (0, Q, 0) \text{ i.e., parallel to the } y\text{-axis} \\ \int_{\mathcal{C}} \vec{F} \cdot d\vec{r} = \int_{\mathcal{C}} R(x, y, z) dz = \int_a^b R(\vec{r}(t)) \frac{dz}{dt} dt & \text{if } \vec{F} = (0, 0, R) \text{ i.e., parallel to the } z\text{-axis} \end{cases}}.$$

Remark 0.3 In practice, to evaluate the integral

$$\int_{\mathcal{C}} \vec{F} \cdot d\vec{r} = \int_{\mathcal{C}} P(x, y, z) dx + Q(x, y, z) dy + R(x, y, z) dz$$

for $\mathcal{C}$ described by the vector function $\vec{r} : [a, b] \rightarrow \mathbb{R}^3$, we replace

$\int_{\mathcal{C}}$	with	$\int_a^b$
x	with	$x(t);$
y	with	$y(t);$
z	with	$z(t);$
dx	with	$\frac{dx}{dt}dt;$
dy	with	$\frac{dy}{dt}dt;$
dz	with	$\frac{dz}{dt}dt.$

We examine some properties of the line integral of a vector function.

Proposition 0.4 *Let $\vec{F}_1, \vec{F}_2 : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ be continuous vector fields. Let $\mathcal{C}$ be a curve contained in D . Then*

$$(a) \quad \int_{\mathcal{C}} (\alpha \vec{F}_1 + \beta \vec{F}_2) \cdot d\vec{r} = \alpha \int_{\mathcal{C}} \vec{F}_1 \cdot d\vec{r} + \beta \int_{\mathcal{C}} \vec{F}_2 \cdot d\vec{r};$$

$$(b) \quad \int_{\mathcal{C}} \vec{F} \cdot d\vec{r} = \int_{\mathcal{C}_1} \vec{F} \cdot d\vec{r} + \int_{\mathcal{C}_2} \vec{F} \cdot d\vec{r}, \text{ where the curves } \mathcal{C}_1 \text{ and } \mathcal{C}_2 \text{ are curves in } D \text{ such that } \mathcal{C} = \mathcal{C}_1 \cup \mathcal{C}_2 \text{ and if } \mathcal{C}_i \text{ is described by the vector function } \vec{r}_i : [a_i, b_i] \rightarrow \mathbb{R}^n \text{ then } \vec{r}_1(b_1) = \vec{r}_2(a_2).$$

0.1.1 Fundamental theorem of calculus for line integrals

We recall from the Fundamental Theorem of Calculus (FTC) that, if $F : [a, b] \rightarrow \mathbb{R}$ is continuously differentiable, then

$$\int_a^b F'(x) dx = F(b) - F(a).$$

The following theorem plays the role of the Fundamental Theorem of Calculus for the line integral of a vector field.

Theorem 0.5 *Let $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be a continuously differentiable function and let $\mathcal{C}$ be a regular curve contained in D and described by the vector function $\vec{r} : [a, b] \rightarrow \mathbb{R}^n$. Then*

$$\boxed{\int_{\mathcal{C}} \nabla f \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))}. \quad (5)$$

Proof: We find that

$$\int_{\mathcal{C}} \nabla f \cdot d\vec{r} = \int_a^b \nabla f(\vec{r}(t)) \cdot \vec{r}'(t) dt \underbrace{=}_{(\text{CR})} \int_a^b \frac{d}{dt} (f \circ \vec{r})(t) dt \underbrace{=}_{(\text{FTC})} f \circ \vec{r}(b) - f \circ \vec{r}(a).$$

■

For example, the work done to move an object by conservative force field $\vec{F}$ from a point A to a point B along a regular curve $\mathcal{C}$ is

$$W = \int_{\mathcal{C}} \vec{F} \cdot d\vec{r} = f(B) - f(A).$$

Clearly W is independent of the choice of path in D from A to B . This is called the **independence of path** for the line integral of a conservative vector field.

Definition 0.6 (Independence of path)

Let $F : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a continuous vector field. We say that the line integral $\int_C \vec{F} \cdot d\vec{r}$ is independent of path if $\int_{C_1} \vec{F} \cdot d\vec{r} = \int_{C_2} \vec{F} \cdot d\vec{r}$ for any two curves C_1 and C_2 contained in D and having the same initial and terminal points.

Remark 0.7 As we have seen above if the vector field $\vec{F}$ is conservative, then its line integral $\int_C \vec{F} \cdot d\vec{r}$ is independent of path. Viceversa, it is also known that under certain conditions on the domain D , if the vector field is such that its line integral $\int_C \vec{F} \cdot d\vec{r}$ is independent of path, then the vector field $\vec{F}$ is conservative. Precisely, The following two facts are equivalent:

- (1) the line integral $\int_C \vec{F} \cdot d\vec{r}$ is independent of path in D ;
- (2) the line integral $\oint_{\mathcal{K}} \vec{F} \cdot d\vec{r} = 0$ for every closed curve $\mathcal{K}$ in D .

Furthermore, it is well-known that, under some conditions on the domain D , (1) and (2) are equivalent to the third fact:

- (3) the vector field $\vec{F}$ is conservative.

Let $\mathcal{C}$ be described by the vector function $\vec{r} : [a, b] \rightarrow \mathbb{R}^n$. Then we denote $-\mathcal{C}$ the curve described by the vector function $\vec{u} : [a, b] \rightarrow \mathbb{R}^n$ defined by

$$\vec{u}(t) := \vec{r}(a + b - t) \quad \forall t \in [a, b].$$

Theorem 0.8 Let $\vec{F} : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with $\vec{F} = (P, Q)$. If $\vec{F}$ is conservative and the functions $P, Q : D \rightarrow \mathbb{R}$ have continuous first order partial derivatives, then

$$\boxed{\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}}. \quad (6)$$

Proof: Let $f : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ be the a function such $\vec{F} = \nabla f$. So, it follows that

$$\begin{cases} f_x = P \\ f_y = Q \end{cases} \Rightarrow \begin{cases} P_y = f_{yx} \\ Q_x = f_{xy} \end{cases}$$

Since P_y and Q_x are continuous by assumption, it follows that f_{xy} and f_{yx} are continuous and hence, from the Clairaut-Schwartz, we obtain that

$$f_{xy} = f_{yx} \quad \Leftrightarrow \quad Q_x = P_y$$

which achieves the proof of the theorem. ■

The next set of notes will be on the Green theorem!